Alireza Fadaei

Full-Stack Software Engineer | Applied AI & Cybersecurity Researcher

Oulu, Finland · Email Alireza · Portfolio · GitHub profile · LinkedIn · ORCID

RESEARCH & ENGINEERING PROFILE

Full-stack software engineer and researcher combining frontend/backend development, data-intensive applications and reliable systems research. I build React and Python/FastAPI software and work on applied AI, federated learning, intrusion detection and distributed systems. Currently pursuing an M.Sc. in Software Engineering and Information Systems at the University of Oulu.

RESEARCH INTERESTS

- Federated & distributed learning
- Privacy-preserving machine learning
- Edge computing & IoT security
- Research software & scientific data platforms
- Post-quantum security integration
- Intrusion detection & cybersecurity
- Reliable AI-enabled systems
- Software architecture & quality
- Applied AI & LLM automation

EDUCATION

M.Sc. Software Engineering and Information Systems University of Oulu, Finland · 2026–Present

M.Sc. Computer Science Islamic Azad University, Najafabad Branch · 2022–2025 · Top student

Thesis: An Optimized Federated Deep Learning Approach for Intrusion Detection in Dew Computing Environments

B.Sc. Computer Engineering University of Isfahan · 2017–2022 · Best B.Sc. Thesis Award

Thesis: Design and Implementation of an ADS-B-Based Aircraft Observation System

PUBLICATION

A Novel Intrusion Detection System for Dew Computing Environments Based on an Enhanced Federated Deep Learning Model Alireza Fadaei, Behrang Barekatain · The Journal of Supercomputing · 2026

Published 16 February 2026 · Read publication (Springer)

Developed a CNN-based federated intrusion-detection approach for resource-constrained distributed environments without centralizing sensitive data.

SELECTED RESEARCH & SOFTWARE PROJECTS

Federated Web Attack Detection — GitHub · Python · TensorFlow · Flower

Implemented a lightweight 1D-CNN federated intrusion-detection model and evaluated IID/non-IID partitioning on Industrial IoT data.

BrassicaLNC — live research database · Full-stack research platform

Developed frontend search/data exploration, backend services, research-data models and server operations for curated biological data from 2,361 RNA-Seq libraries (110 BioProjects).

ML-KEM Modernization — GitHub · University of Oulu course team project · 2026

Designed interfaces between legacy device, edge gateway and cloud service; coordinated five development workstreams using Docker Compose, Git and guidance for AI coding agents.

ADS-B Aircraft Observation Station B.Sc. thesis · Best B.Sc. Thesis Award

Designed a system to capture, process and present live aircraft observations.

PROFESSIONAL EXPERIENCE

Full-Stack Software Engineer | Aminyx (Remote) · 2026

- Developed React interfaces and Python/FastAPI backend services for a modular B2B SaaS platform, integrating application workflows and REST APIs.
- Built AI-driven sourcing and trust-scoring workflows; developed a Python/Pydantic domain-and-contact enrichment pipeline with web crawling, optional LLM resolution, checkpoints and Docker deployment Enrichment pipeline on GitHub.

Full-Stack Software Developer | Fardaad (ISTT) · 2022–2024

- Implemented frontend and backend functionality for vulnerability- and patch-management tools used in municipal and government networks; connected UI, APIs and operational data.
- Improved patch-deployment logic and collaborated through Git-based development and review.

Full-Stack Research Software Developer | BrassicaLNC Project · 2021–2022

- Built searchable frontend views, backend services, research-data models and managed data/platform operations for a scientific database.

Backend Developer & Systems Administrator | MetaMenu · 2021–2022

- Developed backend services and supported cloud-hosted deployment for a restaurant SaaS platform with augmented reality (AR) 3D menus, ordering, delivery and accounting workflows.

Backend Developer | Piranest Team · 2021–2022

- Developed travel marketplace backend features and tour-reservation workflows.

Python Web Developer | Fapsco Project · 2020

- Built and maintained Flask application features and internal business tools.

TEACHING EXPERIENCE

Programming Instructor Iran TVTO · 2024–2026

Delivered practical software-development and programming training.

TECHNICAL SKILLS

Full-stack development React · JavaScript · Python · FastAPI · Node.js · Django · Flask · REST APIs

Data and infrastructure PostgreSQL · SQL · Redis · Neo4j · Docker · Docker Compose · Linux · Nginx · Data modelling

Applied AI and research Pydantic · AI/LLM automation · TensorFlow · Scikit-learn · Flower · Federated learning

Engineering practices Application architecture · Git · Testing · Code review · Technical documentation

LANGUAGES

English: Professional working proficiency (IELTS Academic 7.5, March 2025) · Finnish: Beginner (studying)

BUSINESS APPLICATION DEMOS

CNC Process Management System — live demo

Psychological Clinic CRM — live demo